

Experiment ‘p_5_2_c_sorted’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_2_c_sorted

Natural language statement: Write a method returning true if the arguments are sorted, with the smallest one coming first.

Method signature: p_5_2_c_sorted(x: real, y: real, z: real) returns (is_sorted: bool)

Ensures

- `is_sorted == (x <= y && y <= z)`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method returning true if the arguments are sorted, with the smallest one coming
→ first.

The signature should be:

p_5_2_c_sorted(x: real, y: real, z: real) returns (is_sorted: bool)

The method should respect the following contract:

ensures is_sorted == (x <= y && y <= z)

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmyMgYNY2dTvMrHXxwtQM1pUlHxhp

Model: gpt-5.5-2026-04-23

Created at: 1780564158

Finish reason: stop

Usage: 292 tokens in, and 27 tokens out

```
{  
  is_sorted := x <= y && y <= z;  
}
```

Final Program

```
method p_5_2_c_sorted(x: real, y: real, z: real) returns (is_sorted: bool)  
  ensures is_sorted == (x <= y && y <= z)  
{  
  is_sorted := x <= y && y <= z;  
}
```

Total Token Usage

Input tokens: 292

Output tokens: 27

Reasoning tokens: 0

Sum of ‘total tokens’: 319

Experiment Timings

Overall Experiment started at 1780564157760, ended at 1780564179120, lasting 21360ms (21.36 seconds)

Iteration #1 started at 1780564157766, ended at 1780564179120, lasting 21354ms (21.35 seconds)